

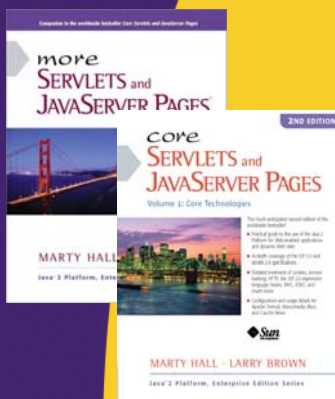


Java: Introduction and Overview

J2EE Training: <http://courses.coreservlets.com/>

Servlets, JSP, Struts, JSF, Hibernate, AJAX, Java 5, etc.

Developed and taught by well-known author and developer. At public venues or onsite at *your* location.



For live Java training, please see training courses at <http://courses.coreservlets.com/>. Servlets, JSP, Struts, JSF, AJAX, Java 5, etc.



Taught by the author of *Core Servlets and JSP*, *More Servlets and JSP*, and this tutorial. Available at public venues, or customized versions can be held on-site at your organization.

Developed and taught by well-known author and developer. At public venues or onsite at *your* location.

Agenda

- **Truths / Myths About Java**
 - Java is Web-Enabled?
 - Java is Safe?
 - Java is Cross-Platform?
 - Java is Simple?
 - Java is Powerful?
- **Common Java Protocols and Packages**
- **Getting Started**
 - Applications
 - Applets

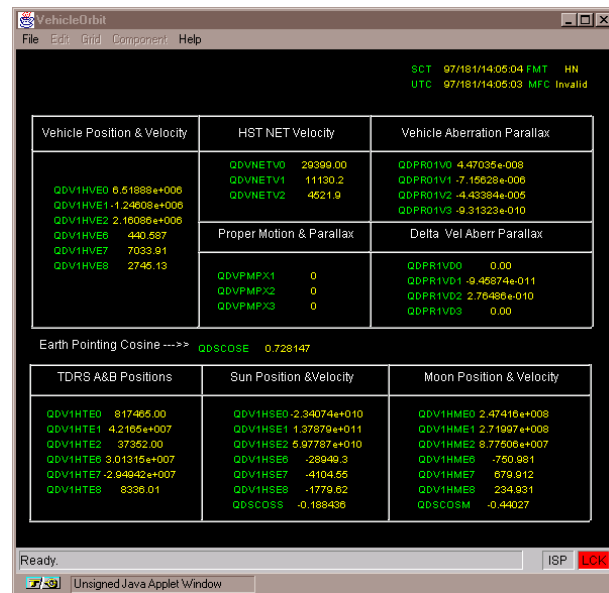
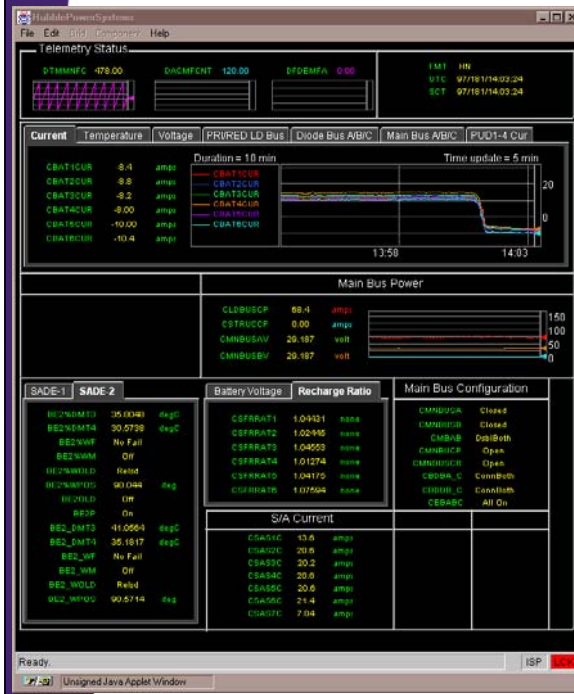
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Java is Web-Enabled?

- **Truth: Web browsers can run Java “applets”**
 - The Web can be used for *software* delivery and *execution*, not just *document* delivery and *display*
 - No more installation or updates; just a bookmark
 - Large, complex applets best suited for intranets. Fits the APL model better than the WWW at large.
- **Truth: Java’s network library is easy to use**
 - Ordinary mortals can do socket programming
 - Standard distributed object protocol and DBMS API

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Hubble Space Telescope Monitoring: “NASA Goddard’s Most Successful SW Project Ever.”



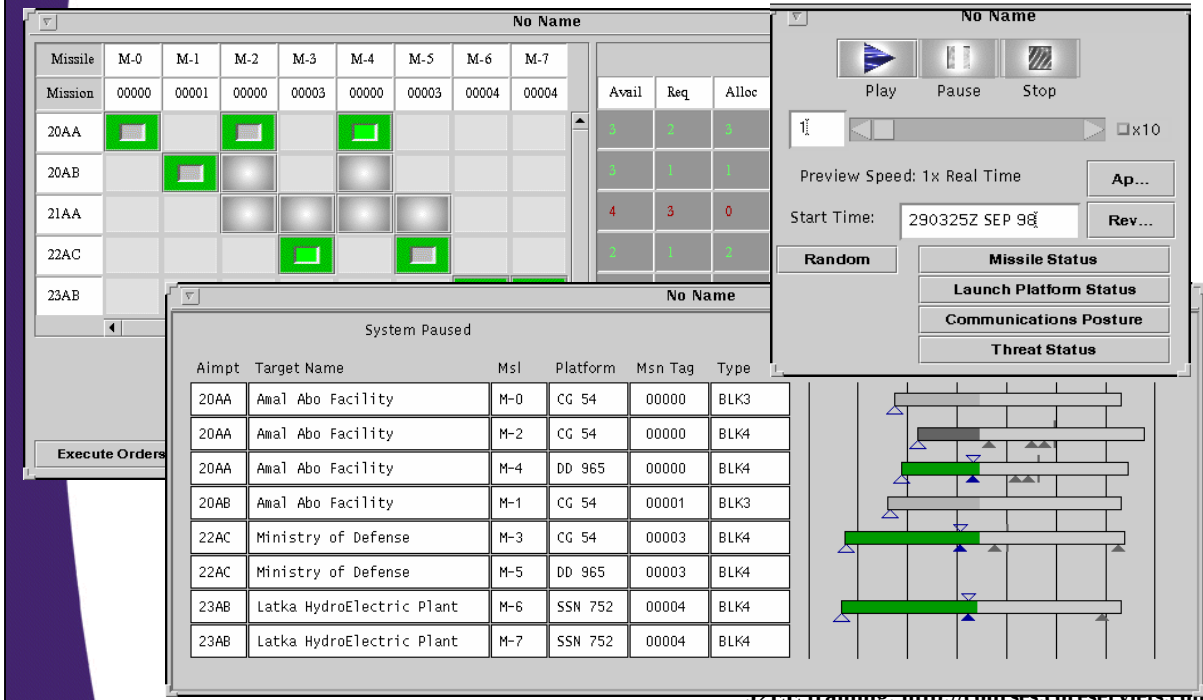
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Java is Web-Enabled?

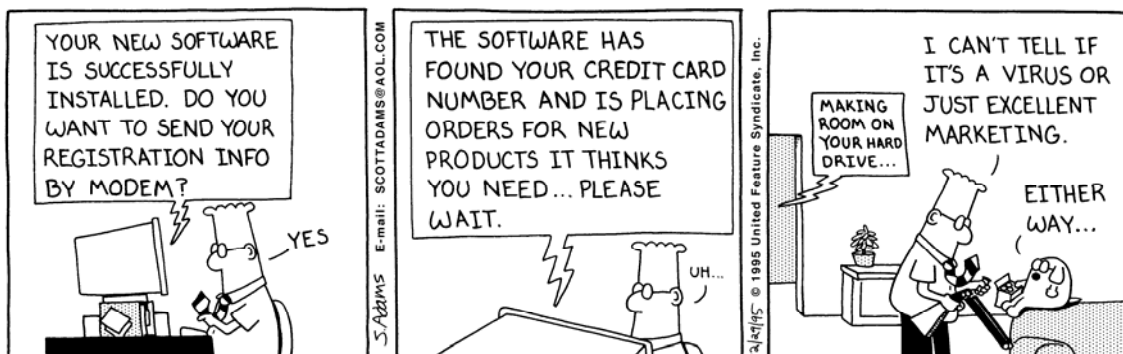
- **Myth: Java is *only* for the Web**
 - Java “applets” run in Web pages
 - Java “applications” run stand-alone
 - Current usage (roughly)
 - Client (applet): 5%
 - Desktop (application): 25%
 - Server (servlets/JSP/EJB): 70%

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Tomahawk Strike Coordination Planner (APL/PPSD)



Java is Safe?



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• JAVA: Just Another Virus Architecture?

Java is Safe?

- **Truth: Restrictions on permissible operations can be enforced**
 - No “raw” memory manipulation (directly or indirectly).
 - Thus, it is easy to identify prohibited operations.
 - Applets, by default, prohibited from:
 - Reading from the local disk
 - Writing to the local disk
 - Executing local programs
 - Opening network connections other than to HTTP server
 - Discovering private info about user (username, directories, OS patch level, applications installed, etc.).

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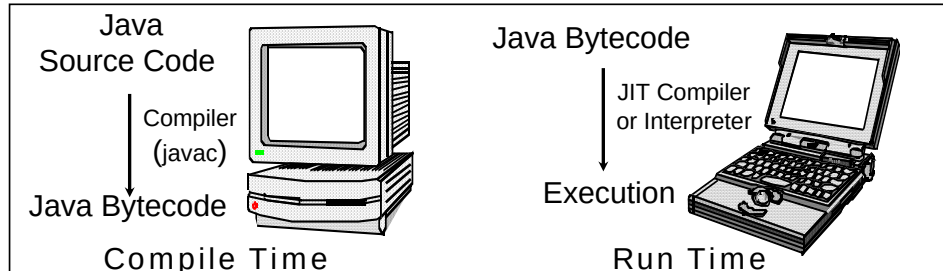
Java is Safe?

- **Myth: Applets cannot harm your computer**
 - Denial of service
 - Browser misconfiguration
 - Implementation bugs
- **Myth: Java is too restricted to be useful**
 - Restrictions apply only to applets, not regular Java programs
 - Digital signatures support relaxed restrictions
- **Myth: Applets with digital signatures are no more or less safe than ActiveX**
 - Relaxed security in applets not “all or nothing” as in ActiveX

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Java is Cross-Platform?

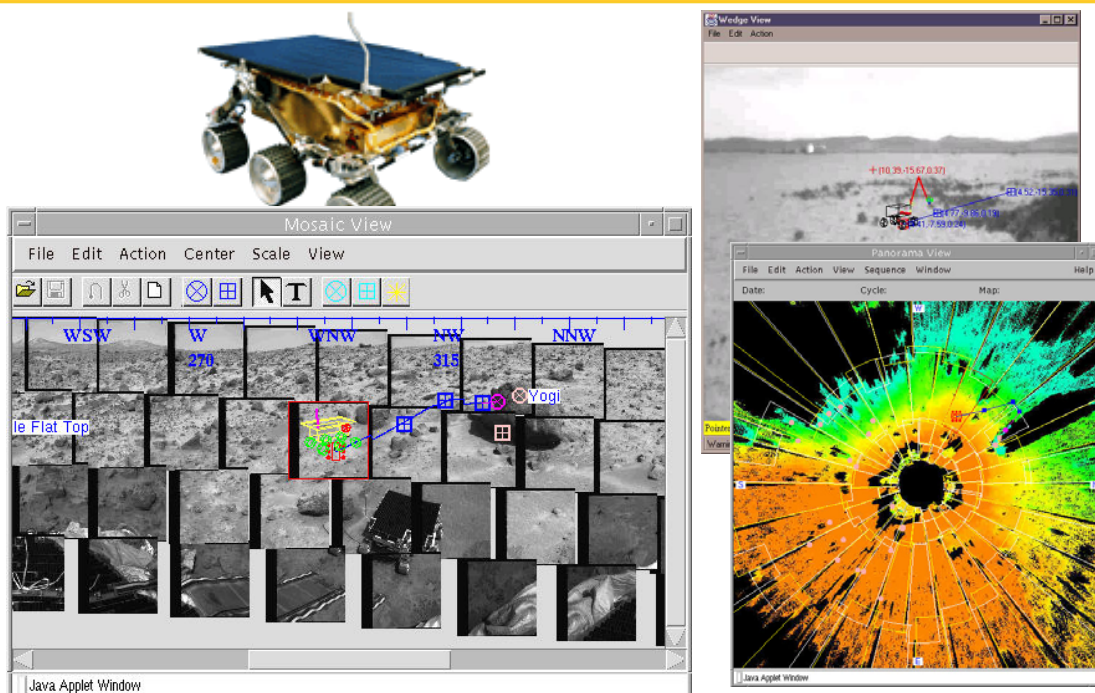
- **Truth:** Java programs can compile to machine-independent bytecode



- **Truth:** All major operating systems have Java runtime environments
 - Most bundle it (Linux, Solaris, MacOS, Windows XP)

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Mars Rover Controller and Simulator



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Java is Cross-Platform?

- **Myth: Safety and machine independence can be achieved with no performance penalty**
 - Current systems are about 20% slower than C++
 - Upcoming releases claim to lower or eliminate that gap
 - I expect the gap to stay at 10% or more
 - Commercial compilers are sometimes faster than free ones
- **Myth: Java is interpreted**
 - Early releases were interpreted
 - Many major “Just in Time” (JIT) compilers

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Java is Cross-Platform?

- **Myth: Write Once Run Anywhere**
 - Cross-platform code can be achieved, but you must test on all platforms you will deliver on.
 - Java apps can execute local code
 - The graphics library behaves slightly differently on different platforms
 - The behavior of the thread scheduler is only loosely defined
- **Myth: Java will kill Microsoft**
 - There is also no longer immediate danger of the reverse (Microsoft killing Java)
 - Microsoft wavered between trying to fight Java and joining it and making money by dominating the market. With .NET, they are back to fighting it again.



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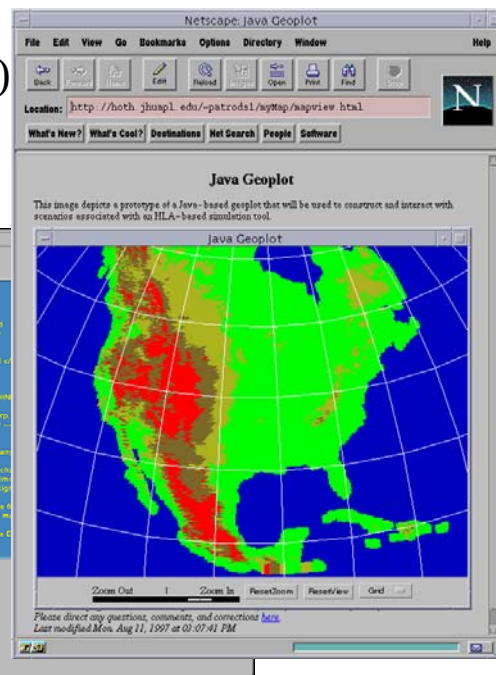
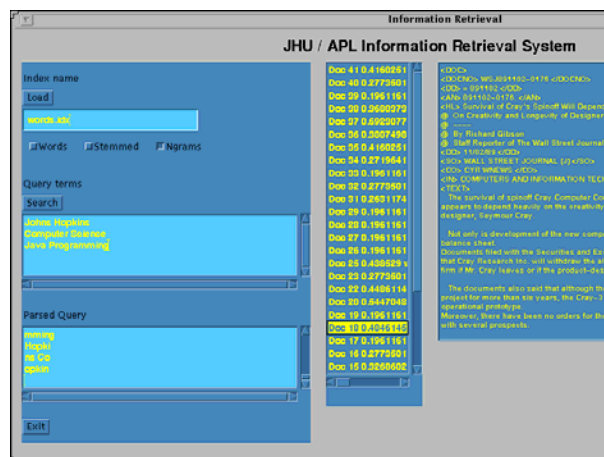
Java is Simple?

- **Truth:** Java greatly simplifies several language features
 - Java has automatic memory management
 - Does Windows and takes out the garbage
 - No dangling pointers. No memory leaks.
 - Java simplifies pointer handling
 - No explicit reference/dereference operations
 - No makefiles for simple applications
 - No header files
 - C++ syntax streamlined
 - C# is comparable to Java, at least as far as the core language goes

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Rapid Application Development in Java

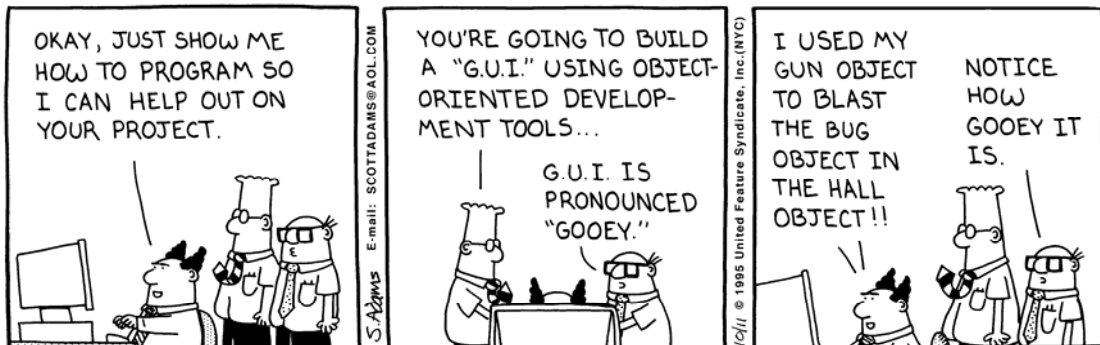
- Information Retrieval for multi-gigabyte text corpus (APL RTDC)
- Geoplot for distributed simulation (APL STD)



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Java is Simple?

- **Myth:** Java programming is simple

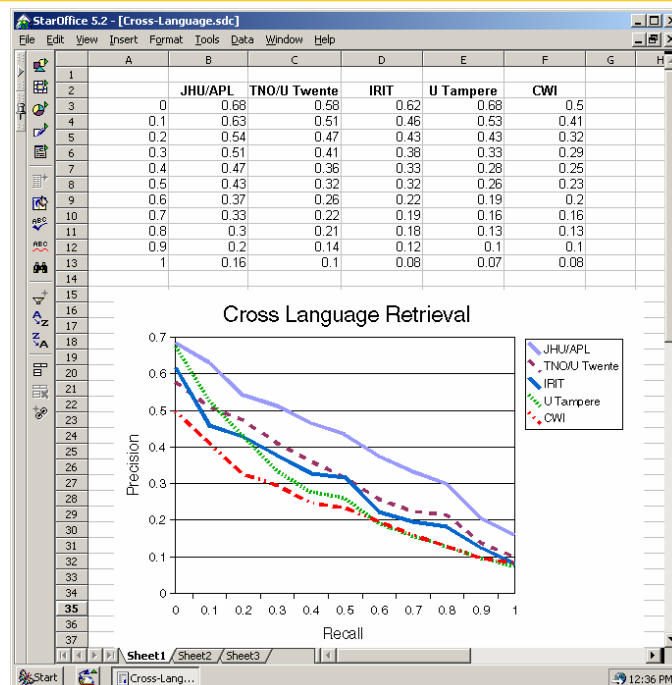


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- Programming is always hard
 - Java is nothing like HTML; only a little bit like JavaScript
- Programmers typically push complexity envelope
 - Multithreaded and network programming

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Star Office MS Office Competitor Written in Java



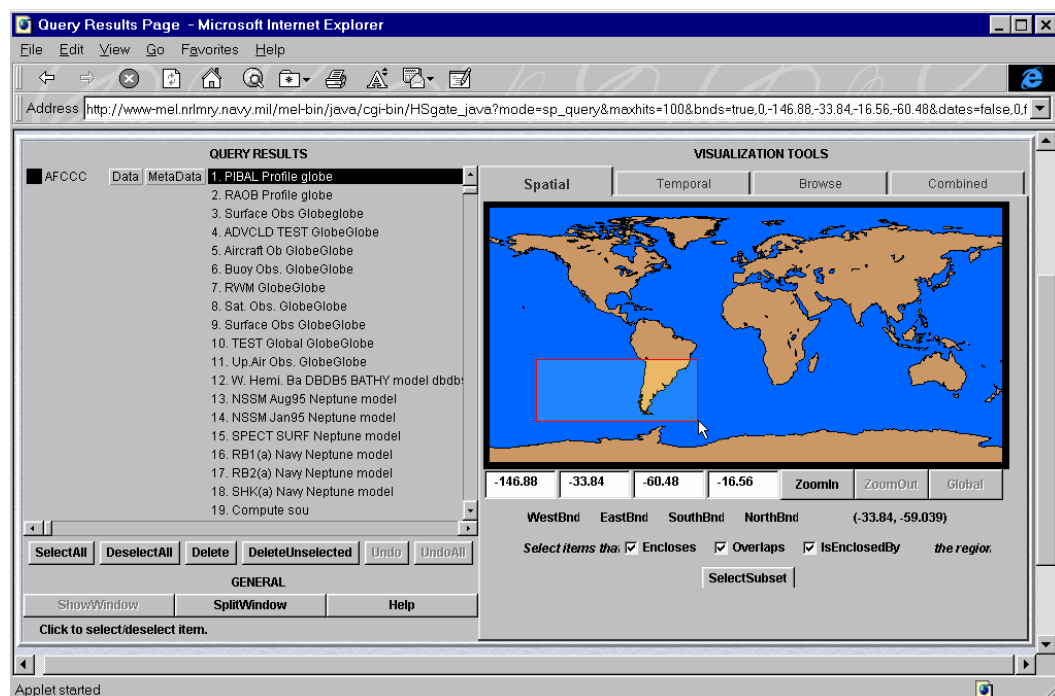
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Java is Powerful?

- **Truth:** Java has a rich set of standard libraries
 - Networking
 - Threads (lightweight processes)
 - Distributed objects
 - Database access
 - Graphics: GUI controls and drawing
 - Data structure library
 - Arbitrary precision integral and fixed-point arithmetic
 - Digital signatures
 - Serialization (transmitting/reassembling data structures)
 - File and stream compression
 - XML parsing

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MEL - Master Environmental Library (DMSO)

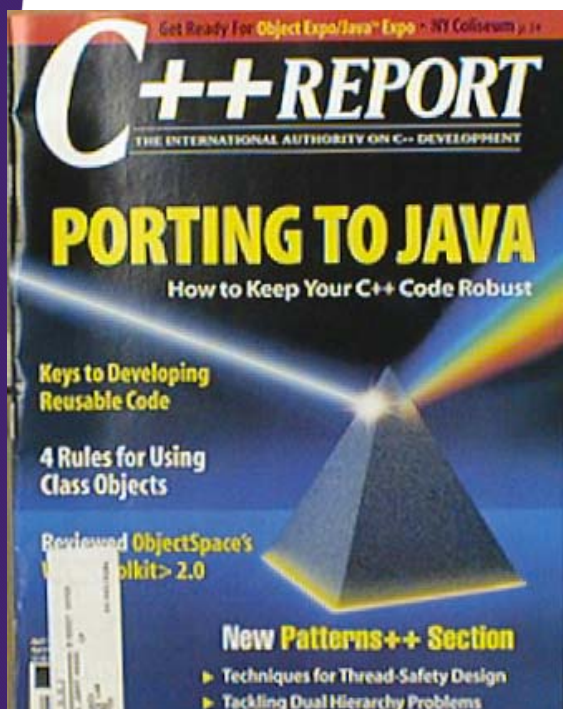


Java is Powerful?

- **Myth:** Java will increase programmer productivity for all applications by XXX%.
- **Myth:** Java will kill C++
- **Myth:** All software should be written in Java
 - Unix utilities: C
 - Small/medium Windows-only programs: Visual Basic
 - String parsing: Perl
 - High-performance, single-platform OO systems: C++
 - Air traffic control, aircraft flight software: Ada
 - Knowledge-based systems: Lisp/CLOS
 - High-performance number crunching: FORTRAN
 - Java also a good alternative for many of these

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Java and C++



Although Java will certainly not kill off C++, Java and C++ do compete for some of the same territory.

Hmm, does *The C++ Report* think that the way to keep your C++ code robust is to port it to Java?

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Key Java Packages and Protocols

- **Core Technologies**

- JDBC
- RMI (and Jini)
- JavaBeans
- Swing
- Java 2D

- **Standard Extensions**

- Servlets
(and JavaServer Pages)
- Enterprise Java Beans (and JNDI)
- Java 3D



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Java Packages and Protocols: JDBC (Java DataBase Connectivity)

- **Standardizes mechanism for making connection to database server**
 - Requires server-specific driver on client. No change to server.
- **Standardizes mechanism for sending queries**
 - Either regular or parameterized queries (stored procedures)
- **Standardizes data structure of query result**
 - Assumes relational data, so data structure is a table
- **Does *not* standardize SQL syntax**
 - Queries are simply strings
 - Server extensions and enhancements supported

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Java Packages and Protocols: Remote Method Invocation (RMI)

- **Built-in Distributed Object Protocol**
 - RMI lets a developer access a Java object and manipulate it in the normal manner. Behind the scenes, each function call really goes over the network to a remote object.
 - Arbitrary Java data structures can be sent over the network with little or no special packaging, due to Java's "serialization" mechanism
 - Similar to a simplified CORBA, but restricted to Java-to-Java communication
- **Jini**
 - RMI-based protocol for self-documenting services.
 - Allows real "plug and play" -- no separate drivers
 - Jury is out on eventual success. Security and industry adoption are open questions



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Java Packages and Protocols: JavaBeans

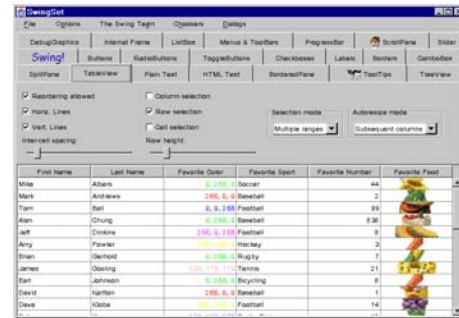
- **JavaBeans is to Java as ActiveX is to Visual C++.**
 - Lets you package a Java program as a software "component"
 - Visual tools can modify/manipulate it without knowing anything about it in advance
 - For example, you can drop a Bean into Visual Café, IBM VisualAge for Java, Inprise (Borland) JBuilder, Sybase PowerJ, Metrowerks CodeWarrior, Sun JavaWorkshop, etc., and it is automatically available from their tool palette for drag-and-drop development
 - Better security and portability than ActiveX
 - More ActiveX components available



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Java Packages and Protocols: Swing

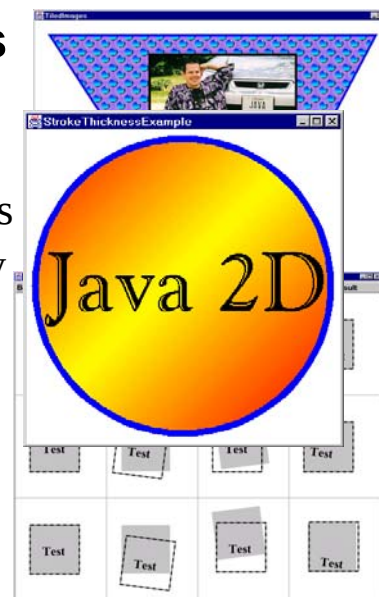
- **Standard GUI-control (widget) library in Java 2**
- **Many more built-in controls**
- **Much more flexible and customizable**
- **Includes many small features aimed at commercial applications**
 - Tooltips, tabbed panes, on-line help, HTML support, dockable toolbars, multi-document interface, etc.
- **Look and feel can be changed at run time**



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Java Packages and Protocols: Java 2D

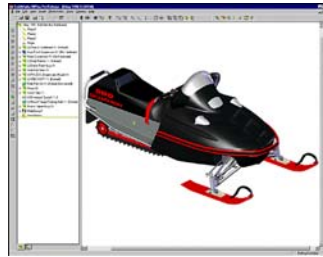
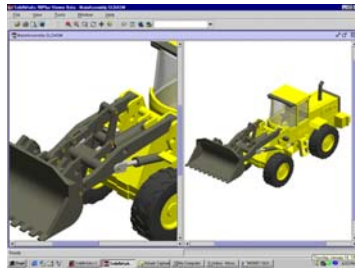
- **Standard drawing library in Java 2**
- **Many new drawing attributes**
 - Fill patterns and images
 - Arbitrary fonts
 - Pen thicknesses and dashing patterns
 - Color mixing rules and transparency
- **Coordinate transformations**
 - Floating-point coordinate system
 - Mapping from memory coords to screen or printer coords
 - Affine transforms: translate, scale, rotate, and shear



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Java Packages and Protocols: Java 3D

- **Standard extension to Java**
 - Not part of “core” Java language like Java 2D
- **Built on top of Direct3D or OpenGL, depending on platform**
- **Scene-graph based model, not primarily immediate-mode rendering**



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Java Packages and Protocols: Servlets and JavaServer Pages (JSP)

- **Servlets: Java's answer to CGI**
 - Efficient: thread, not process, per request
 - Convenient: HTTP headers, cookies, etc.
 - Powerful: persistence, session tracking, etc.
 - Secure: no buffer overflows or shell escapes
- **JavaServer Pages (JSP)**
 - Convenient and efficient way to combine servlets and HTML. Portable alternative to ASP & ColdFusion.
- **Very widely supported:**
 - Supported by servers from Apache, IBM, BEA, Oracle, etc.
 - Used in Web sites of ebay.com, travelocity.com, walmart.com, orbitz.com, hotels.com, NY stock exchange, sears.com, IRS, GSA, Army, Navy, CIA, NSA, etc.

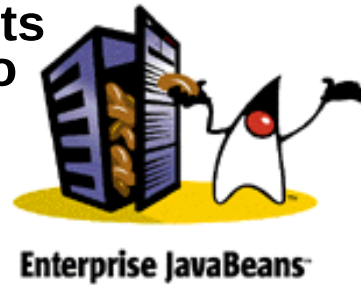


JAVASERVER PAGES™
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Java Packages and Protocols: Enterprise JavaBeans (EJB)

- EJBs are to server components what regular JavaBeans are to application components
- Standardizes access to services like load balancing, persistence, failover, etc.
- Builds on JavaBeans, CORBA, and RMI “under the hood”
- Potentially accessible via non-Java programs
- **Application Servers Supporting EJB**
 - BEA WebLogic, IBM WebSphere, Netscape/iPlanet, Oracle, Progress SW Apptivity, NetDynamics, Sybase, Bluestone, Saphire/Web etc.



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The Future of Java: Java is Driving the Software Industry



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Getting Started

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Servlets, JSP, Struts, JSF, Hibernate, AJAX, Java 5, etc.

Developed and taught by well-known author and developer. At public venues or onsite at *your* location.

Which Java Version Should You Use?

- **Applets**
 - Java Plug-In is required if you want to use Java 2 (1.2 or later) on a browser other than Netscape 6 or 7. Mozilla Firefox depends on the Java plug-in.
 - Most Java applet sites (e.g., *Yahoo! Games*) use Java 1.4
 - Use JDK 1.1 (!) if you want to support WWW at large.
- **Desktop Applications**
 - For best portability to existing deployments, use JDK 1.4
 - For best power and speed, use JDK 5 (aka 1.5)
 - If you install, use 1.5
- **Server-Side Applications**
 - Use 1.4 or 1.5, depending on what existing servers use
 - If you can choose, use 1.5

Getting Started: Nuts and Bolts

- **Install Java**

- JDK 5: <http://java.sun.com/j2se/1.5.0/download.jsp>
- JDK 1.4: <http://java.sun.com/j2se/1.4.2/download.html>
 - Get J2SE (Standard Edition), not J2EE (Enterprise Edition)
 - Get JDK, not just JRE
 - Do not get version bundled with NetBeans IDE
 - If you install after installing browser, system will install plugin automatically.

- **Bookmark or install the Java Documentation**

- <http://java.sun.com/j2se/1.5.0/docs/api/>
- <http://java.sun.com/j2se/1.4.2/docs/api/>

- **All links above**

- <http://www.apl.jhu.edu/~hall/java/>
 - Also links to tutorials, editors, IDEs, etc.

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Creating and Running Program: Manual Approach

- **Create the File**

- Write and save a file (say **Test.java**) that defines public class **Test**
- File and class names are case sensitive

- **Compile the program**

- Compile Test.java through
 - > **javac Test.java**
 - This step creates a file called Test.class
- If you get a “deprecation” warning, this means you are using a Java construct that has a newer alternative (i.e., it still works but is not recommended)
 - Use “javac -deprecation Test.java” for an explanation, then look the newer construct up in the on-line API

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Creating and Running Program: Manual Approach (Continued)

- **Run the program**

- For a stand-alone application, run it with

> java Test

- Note that the command is **java**, not **javac**, and that you refer to **Test**, not **Test.class**
 - To run applets
 - Load the HTML page that refers to it

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Using a Smart Editor or IDE

- **Manual**

- Create file
 - Notepad or any editor
 - Compile file
 - Open DOS window
 - Type "javac MyClass.java"
 - Execute program
 - Open DOS window
 - Type "java MyClass"

- **Editor or IDE**

- Create file
 - Editor such as TextPad, UltraEdit, or jEdit
 - IDE such as Eclipse, IDEA, JBuilder, or NetBeans
 - Compile file
 - Build/Make/Compile from within editor
 - Execute program
 - Run from within editor

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Basic Hello World Application

- “Application” is Java lingo for a stand-alone Java program
 - Note that the class name and the filename *must* match
 - A file can contain multiple classes, but only one can be declared public, and that one’s name must match the filename

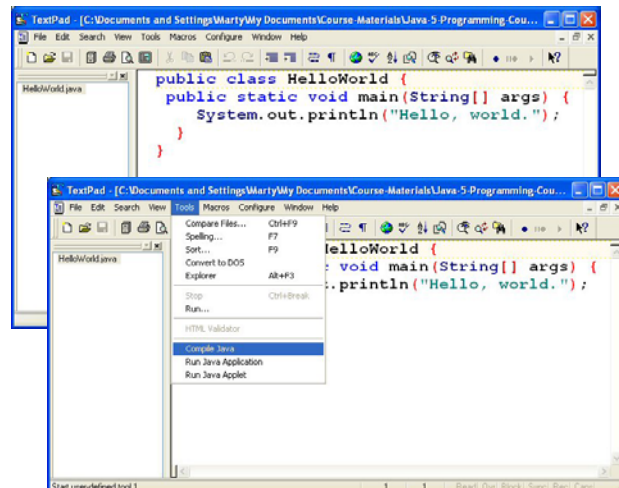
- File HelloWorld.java:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, world.");  
    }  
}
```

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Basic Hello World Application (Continued)

- Compiling:
javac HelloWorld.java
- Running:
java HelloWorld
- Output:
Hello, world.



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Command Line Arguments

- **File ShowArgs.java:**

```
public class ShowArgs {  
    public static void main(String[] args) {  
        for(int i=0; i<args.length; i++) {  
            System.out.println("Arg " + i + " is "  
                               + args[i]);  
        }  
    }  
}
```

- **Differences from C**

- In Java, String is a real type
- Java arrays have an associated length
- The filename is not part of the command line arguments

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Command Line Arguments, Results

- **Compiling and Running:**

```
> javac ShowArgs.java
```

```
> java ShowArgs fee fie foe fum
```

```
Arg 0 is fee
```

```
Arg 1 is fie
```

```
Arg 2 is foe
```

```
Arg 3 is fum
```

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Command Line Arguments: the for/each Loop

- **File ShowArgs2.java:**

```
public class ShowArgs2 {  
    public static void main(String[] args) {  
        for(String arg: args) {  
            System.out.println("Command line arg: "  
                               + arg);  
        }  
    }  
}
```

- **Differences from C**

- New loop construct

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Command Line Arguments, Results

- **Compiling and Running:**

```
> javac ShowArgs2.java
```

```
> java ShowArgs2 fee fie foe fum
```

```
Command line arg: fee
```

```
Command line arg: fie
```

```
Command line arg: foe
```

```
Command line arg: fum
```

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Basic Hello WWW Applet

- File HelloWorld.java:

```
import java.applet.Applet;
import java.awt.*;

public class HelloWorld extends Applet {
    public void init() {
        setBackground(Color.gray);
        setForeground(Color.white);
        setFont(new Font("SansSerif", Font.BOLD, 30));
    }

    public void paint(Graphics g) {
        g.drawString("Hello, World Wide Web.", 5, 35);
    }
}
```

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Basic Hello WWW Applet (Continued)

- File HelloWorld.html:

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0
    Transitional//EN">
<HTML>
<HEAD>
    <TITLE>HelloWW: Simple Applet Test.</TITLE>
</HEAD>

<BODY>
<H1>HelloWW: Simple Applet Test.</H1>

<APPLET CODE="HelloWW.class" WIDTH=400 HEIGHT=40>
    <B>Error! You must use a Java enabled browser.</B>
</APPLET>

</BODY>
</HTML>
```

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Basic Hello WWW Applet (Continued)

- **Compiling:**
`javac HelloWorld.java`
- **Running:**
Load `HelloWW.html` in a Java-enabled browser



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Summary

- **Java is a general purpose language**
 - Supporting standalone applications, browser-based applets, and server-side programs (servlets, etc.)
- **Java is compiled to bytecode**
 - Which can be run on any platform that supports a Java Virtual Machine
- **Java 2 Platform has multiple editions**
 - Standard Edition, Enterprise Edition, or Micro Edition
- **Compiling: use “javac”**
- **Executing standalone programs: use “java”**
- **Executing applets: load HTML file in browser**

– Browsers require Java plug-in J2EE training: <http://courses.coreservlets.com>



Questions?

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Servlets, JSP, Struts, JSF, Hibernate, AJAX, Java 5, etc.

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